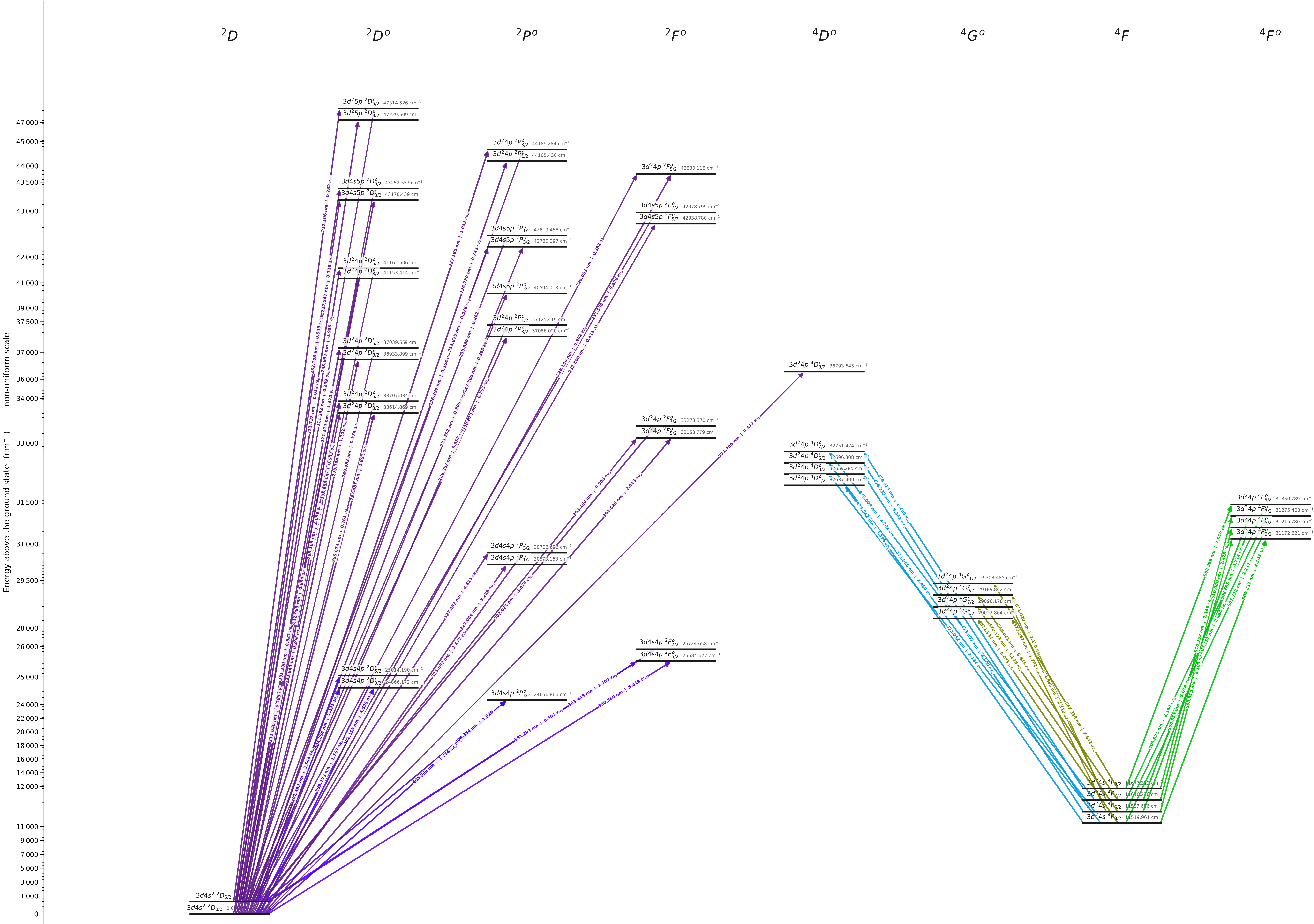
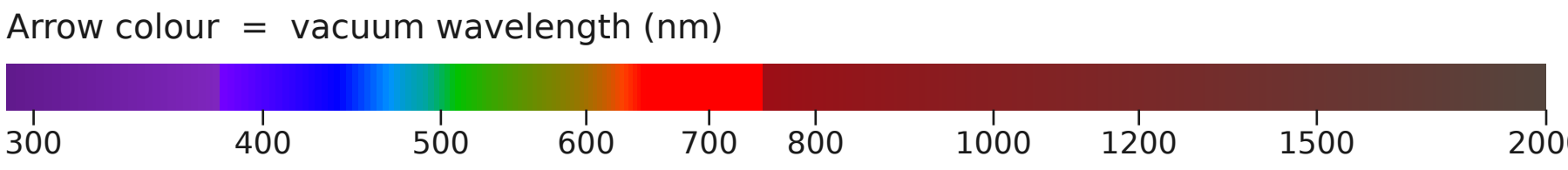


Scandium

Sc (Sc I). Strongest classified transitions below 2 μm (NIST): 49 levels, 70 lines.

Sc+ ionisation limit 52922.00 cm⁻¹ = 6.56149 eV (188.957 nm)

Arrow label: vacuum excitation | reduced dipole matrix element $\langle U||er||J' \rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d4s ² ² D _{3/2 - 3d²4p ²F_{5/2}}	301.6247	301.5369	993.9253	2.518	7.800e+07	-	-	
3d4s ² ² D _{3/2 - 3d²4p ²F_{7/2}}	302.0233	301.9353	992.6138	3.076	8.700e+07	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²P_{1/2}}	327.0842	326.9900	916.5604	3.288	3.130e+08	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²P_{3/2}}	327.4570	327.3627	915.5170	4.413	2.810e+08	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²F_{5/2}}	390.8597	390.7490	767.0078	5.418	1.660e+08	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²F_{7/2}}	391.2926	391.1818	766.1592	6.507	1.790e+08	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²D_{3/2}}	402.1528	402.0391	745.4691	4.575	1.630e+08	-	-	
3d4s ² ² D _{3/2 - 3d4s4p ²D_{5/2}}	402.4817	402.3679	744.8599	5.644	1.650e+08	-	-	
3d ² 4s ⁴ F _{3/2} - 3d ² 4p ⁴ D _{1/2}	473.5421	473.4096	633.0852	3.396	1.100e+08	-	-	
3d ² 4s ⁴ F _{3/2} - 3d ² 4p ⁴ D _{3/2}	473.8971	473.7645	632.6109	4.300	8.800e+07	-	-	
3d ² 4s ⁴ F _{7/2} - 3d ² 4p ⁴ D _{5/2}	474.2354	474.1028	632.1595	5.361	9.100e+07	-	-	
3d ² 4s ⁴ F _{3/2} - 3d ² 4p ⁴ D _{3/2}	474.5147	474.3820	631.7875	6.430	9.800e+07	-	-	
3d ² 4s ⁴ F _{9/2} - 3d ² 4p ⁴ F _{5/2}	508.2986	508.1569	589.7960	7.019	7.600e+07	-	-	
3d ² 4s ⁴ F _{7/2} - 3d ² 4p ⁴ F _{7/2}	508.5134	508.3717	589.5468	5.674	6.200e+07	-	-	
3d ² 4s ⁴ F _{3/2} - 3d ² 4p ⁴ F _{3/2}	508.6955	508.5538	589.3357	4.714	5.700e+07	-	-	
3d ² 4s ⁴ F _{3/2} - 3d ² 4p ⁴ F _{5/2}	508.8370	508.6952	589.1719	4.143	6.600e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).